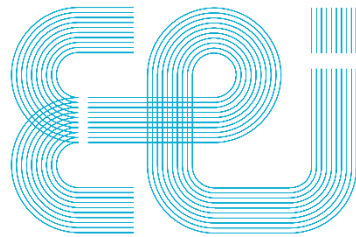


PROGRAMACIÓN TwinCAT

Lab 03

Automatización Industrial
Grado Ingeniería Industrial y Automática

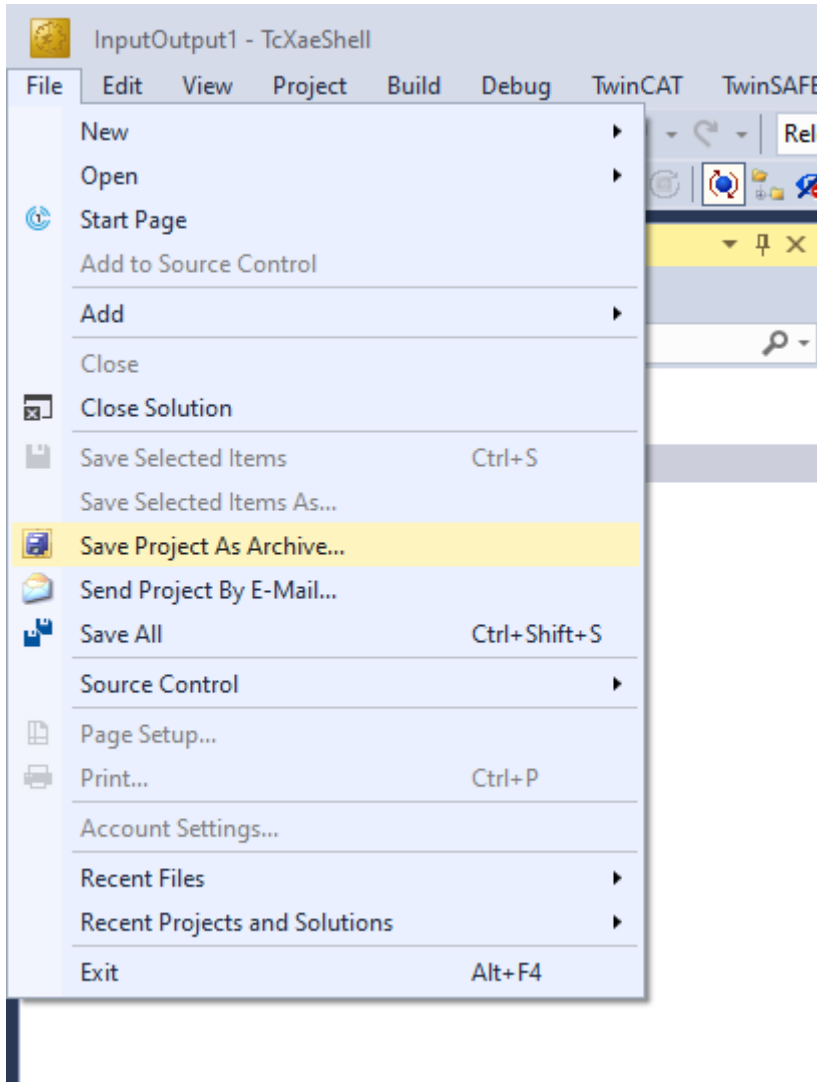


El contenido de esta presentación está basado en los materiales ofrecidos por
Jakob Sagatowski en <https://www.sagatowski.com/>

ÍNDICE

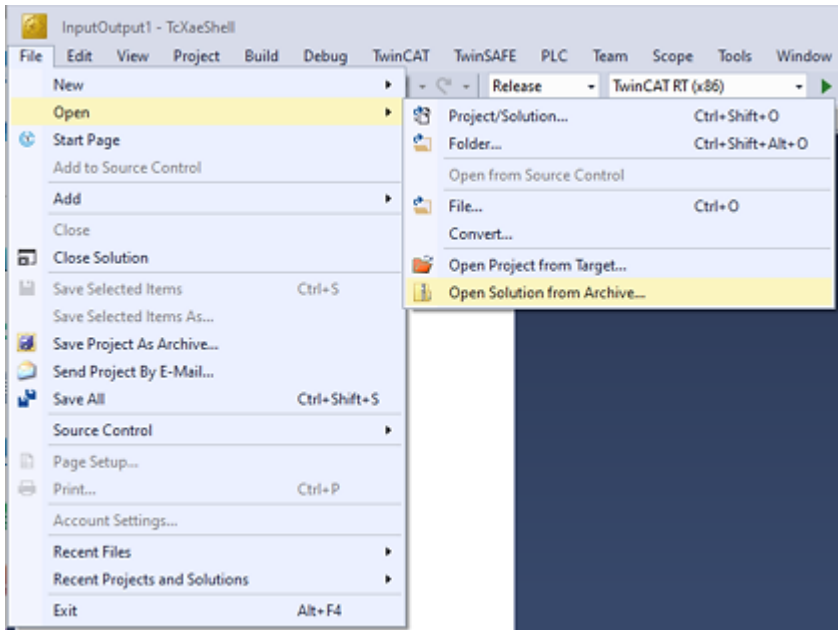
- Cargar y Guardar Proyectos
- Autómata Finito
- Input/Output

Guardar



- Guarda proyecto en formato “tnzip”
- Es fichero comprimido con toda información

Cargar



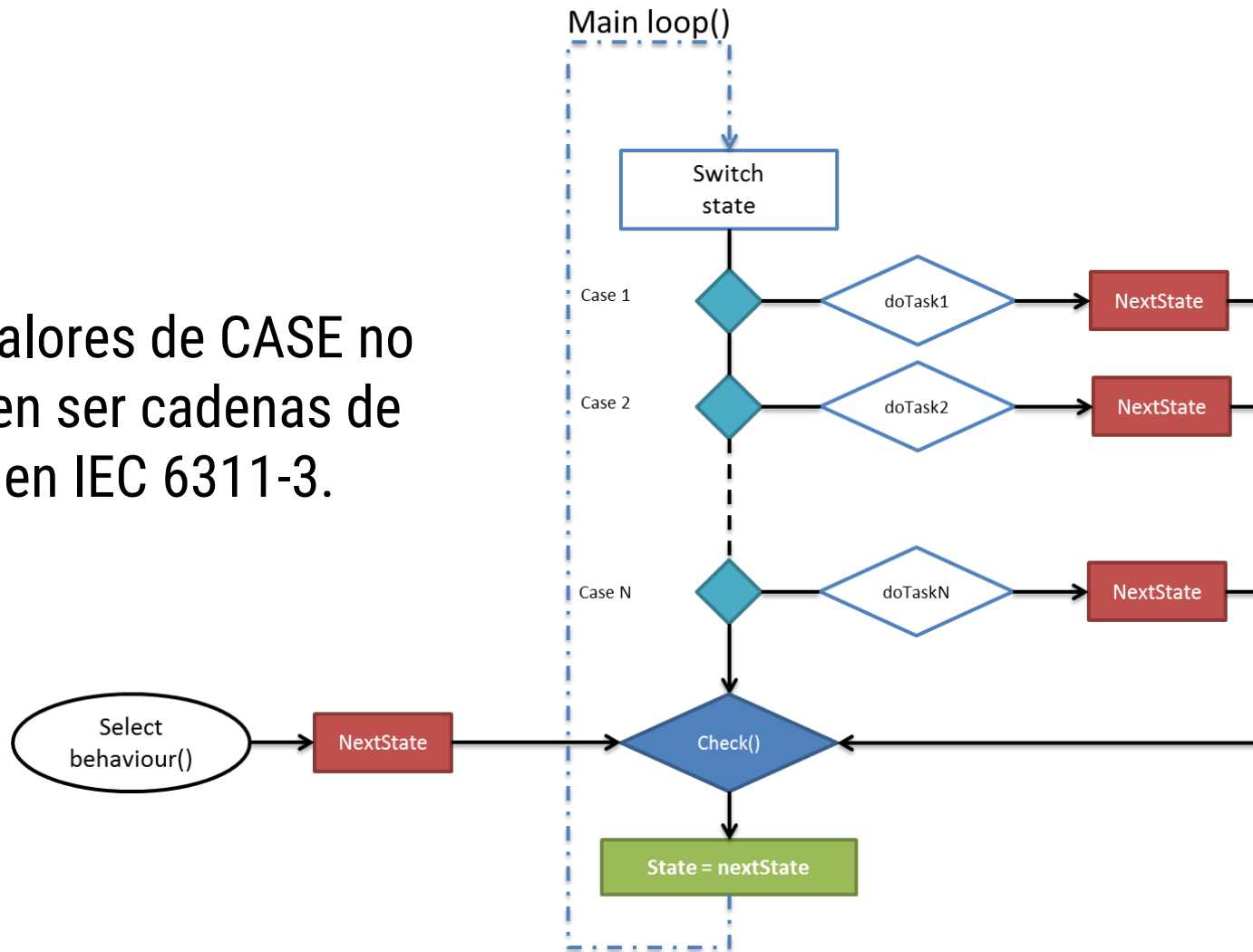
- Carga fichero “tnzip”
- Solicita directorio destino para descomprimir

ÍNDICE

- Cargar y Salvar Proyectos
- **Autómata Finito**
- Input/Output

Autómata Finito

Los valores de CASE no pueden ser cadenas de texto en IEC 6311-3.



Autómata Finito

```
FUNCTION_BLOCK FB_TextFileWriter
VAR_IN_OUT CONSTANT
    sTextToBeWritten : Tc2_System.T_MaxString;
END_VAR
VAR
    nTextWriteState : INT;
END_VAR
-----
CASE nTextWriteState OF

    10 : // Open file
        // Code

    20 : // Write "sTextToBeWritten" to file
        // Code

    30 : // Close file
        // Code

    40 : // Error handling
        // Code

END_CASE
```

```
FUNCTION_BLOCK FB_TextFileWriter
VAR_IN_OUT CONSTANT
    sTextToBeWritten : Tc2_System.T_MaxString;
END_VAR
VAR
    eTextWriteState : (OPEN_FILE, WRITE_TO_FILE, CLOSE_FILE, ERROR);
END_VAR
-----
CASE eTextWriteState OF

    OPEN_FILE :
        // Code

    WRITE_TO_FILE :
        // Code

    CLOSE_FILE :
        // Code

    ERROR :
        // Code

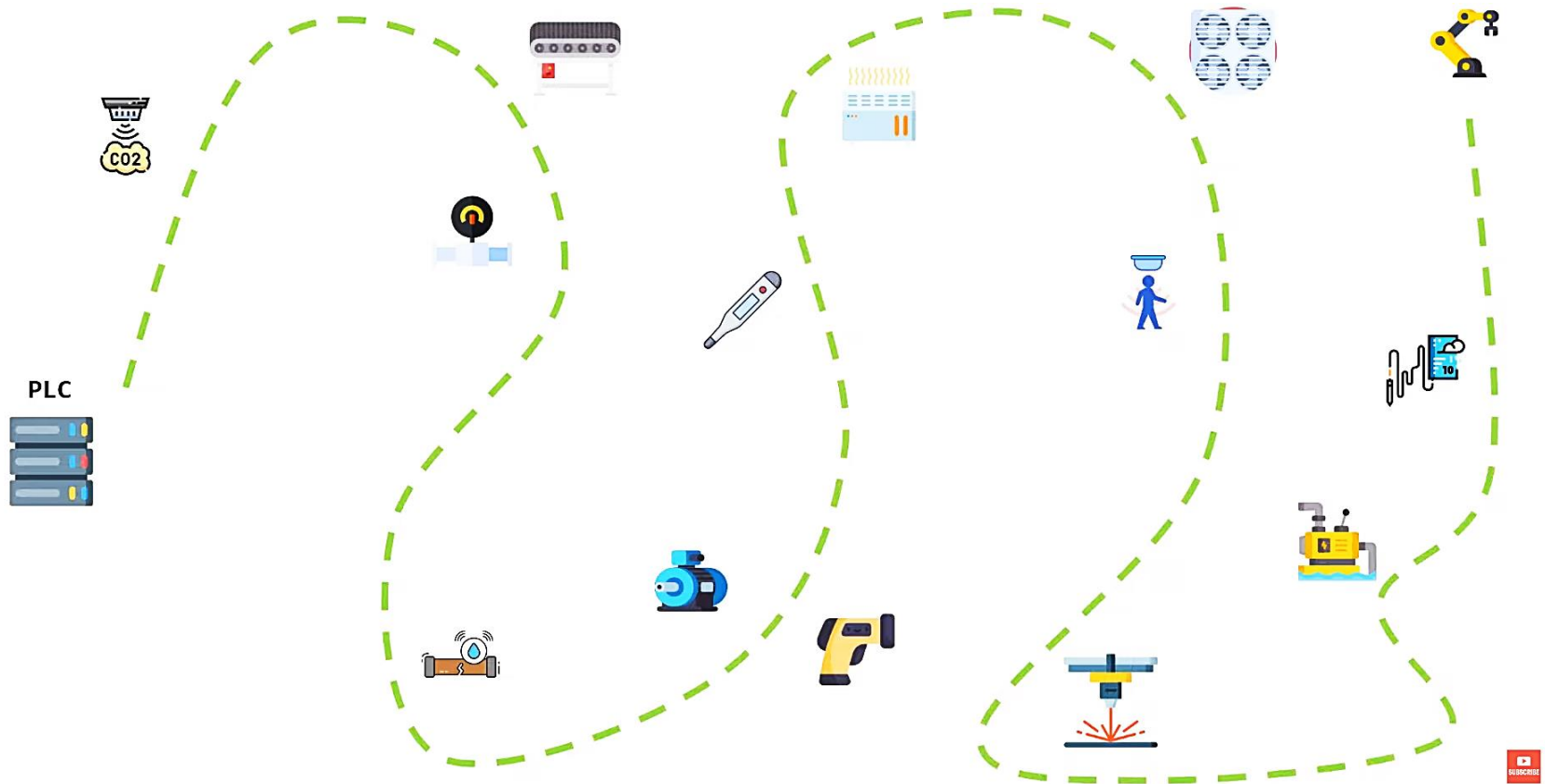
END_CASE
```

Uso enumeraciones locales
mejora legibilidad.

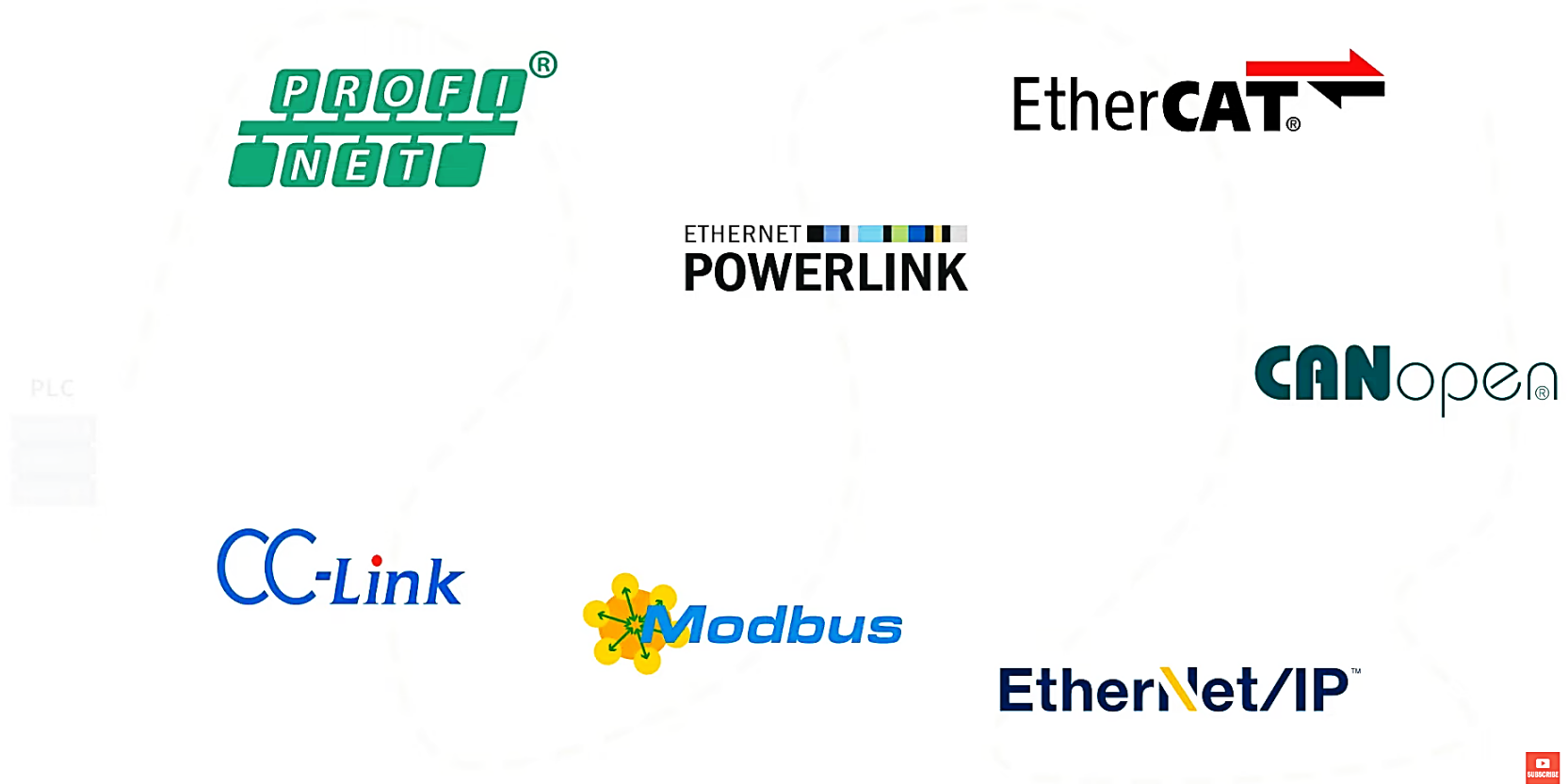
ÍNDICE

- Cargar y Salvar Proyectos
- Autómata Finito
- **Input/Output**

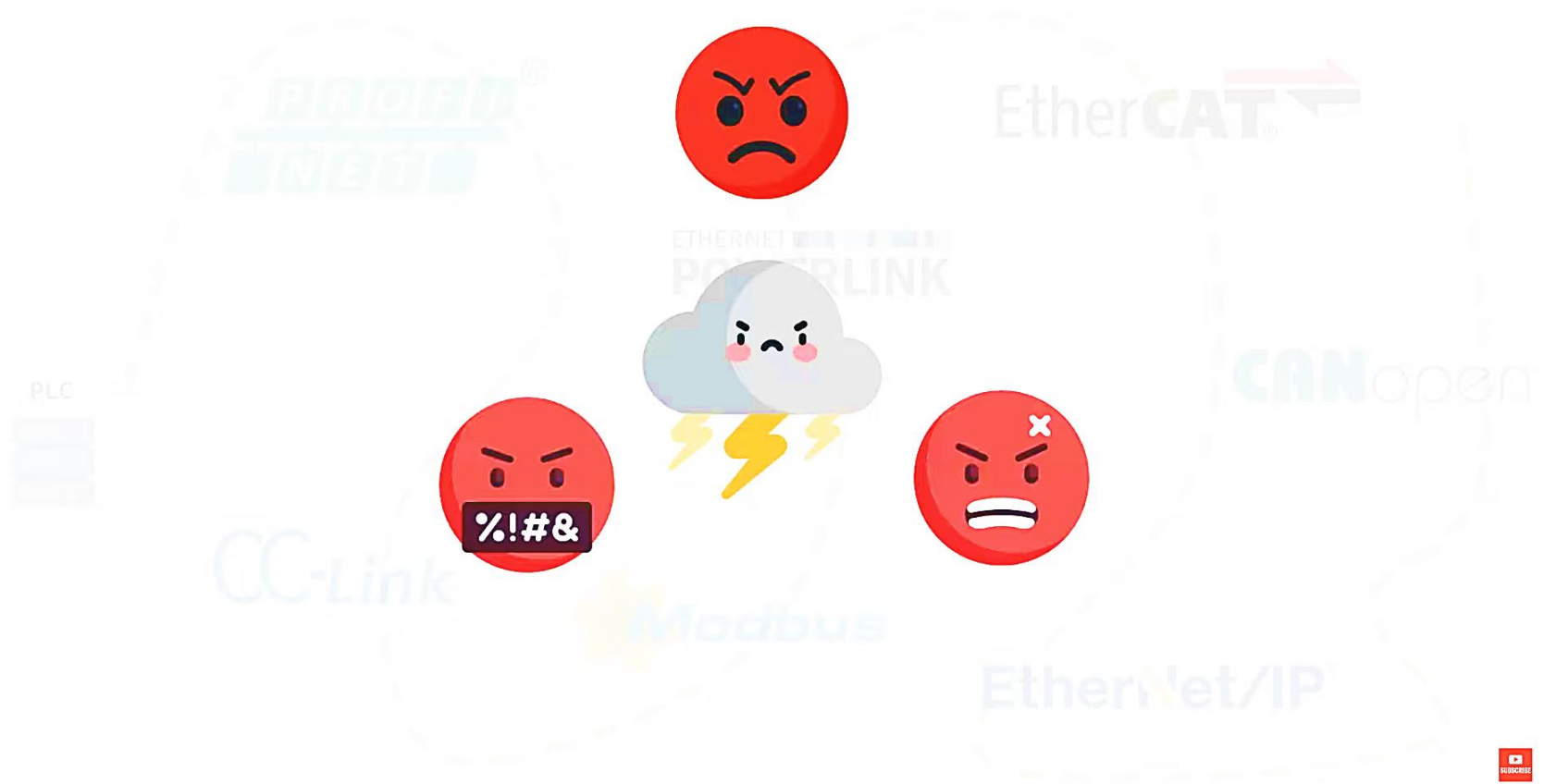
Input/Output



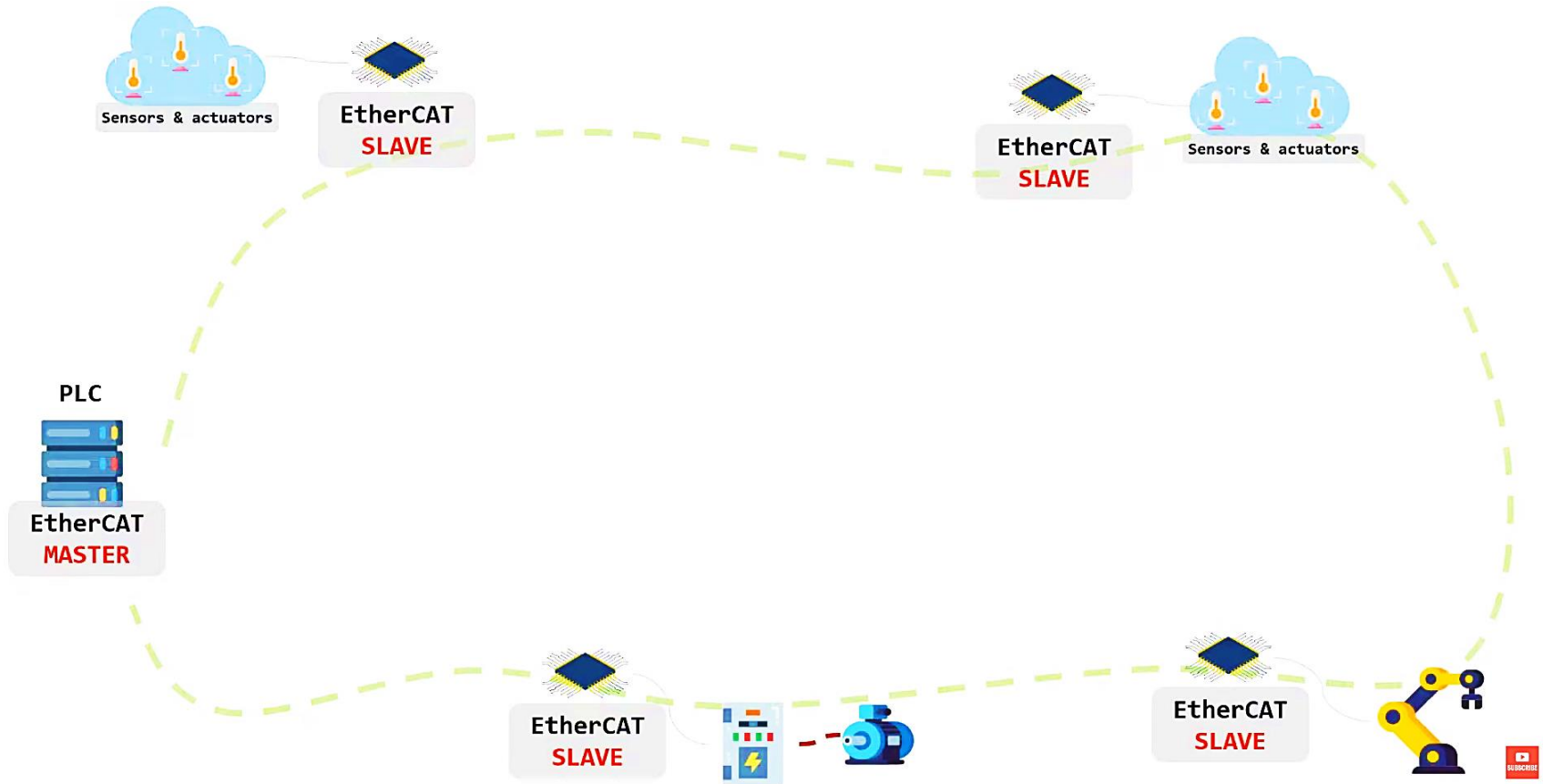
Bus de Campo



Bus de Campo

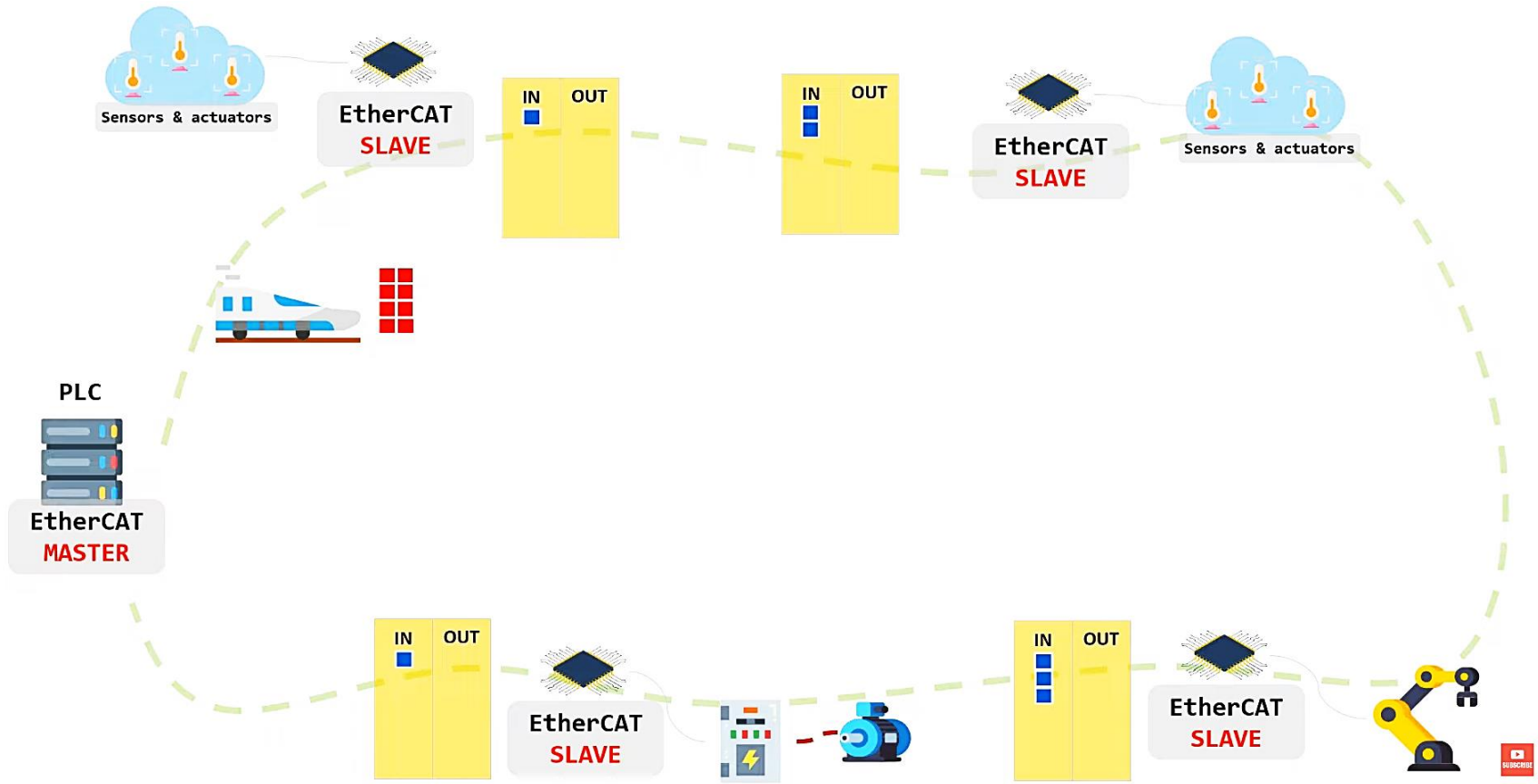


EtherCAT



MASTER = envía mensajes capturados por todos SLAVE

EtherCAT

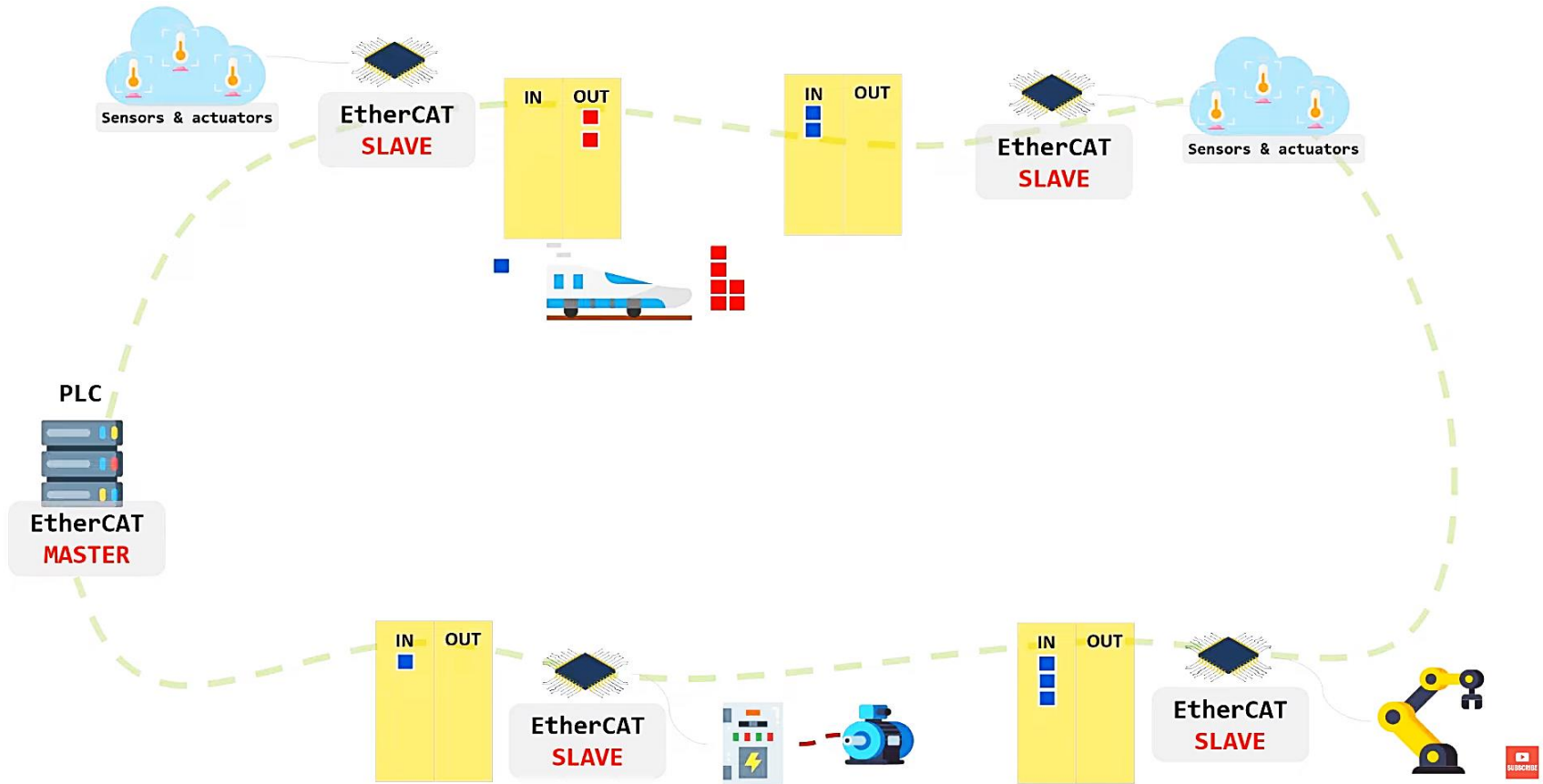


IN = datos slave > master

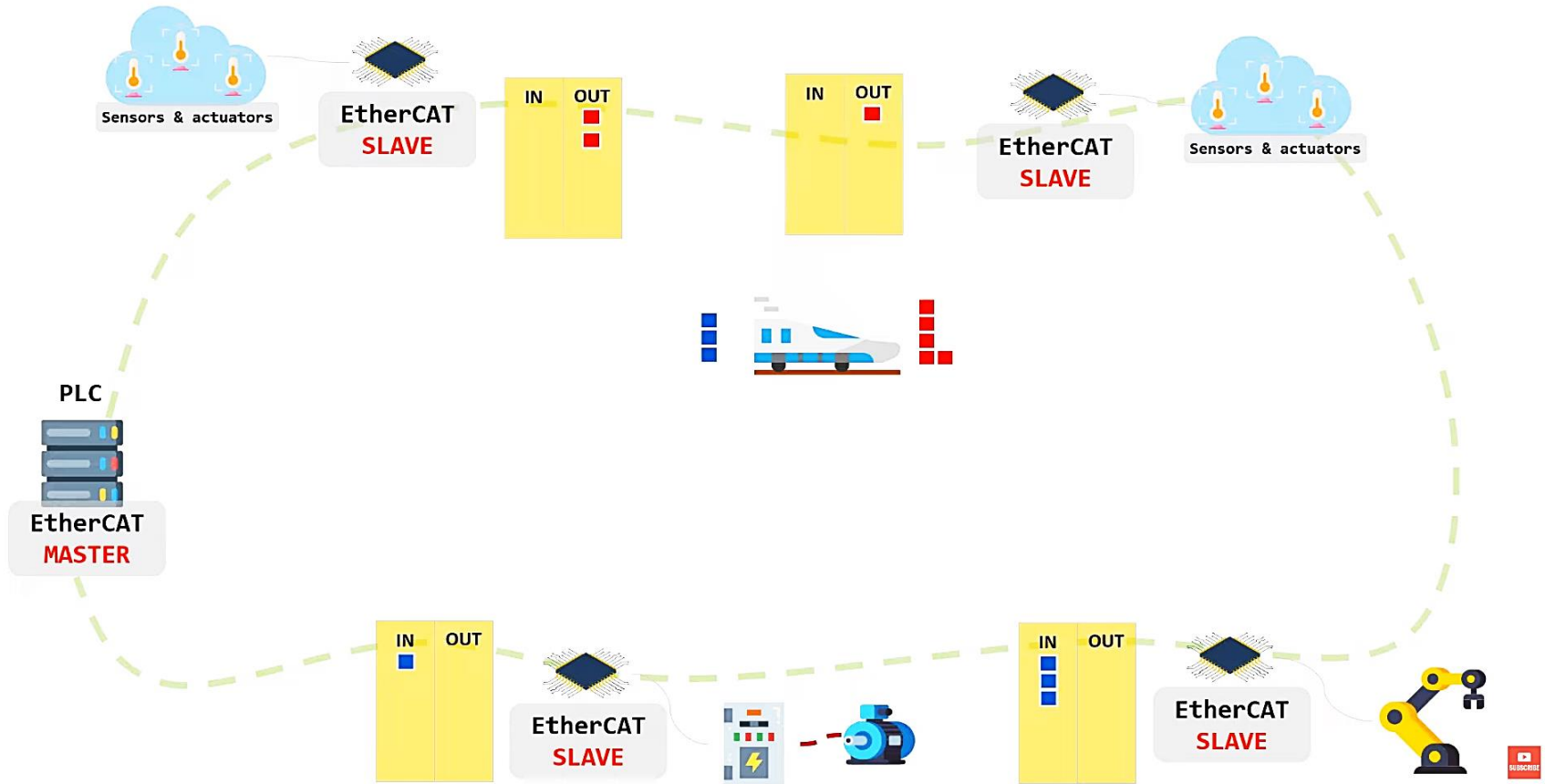
OUT = datos master > slave

criterio punto vista MASTER

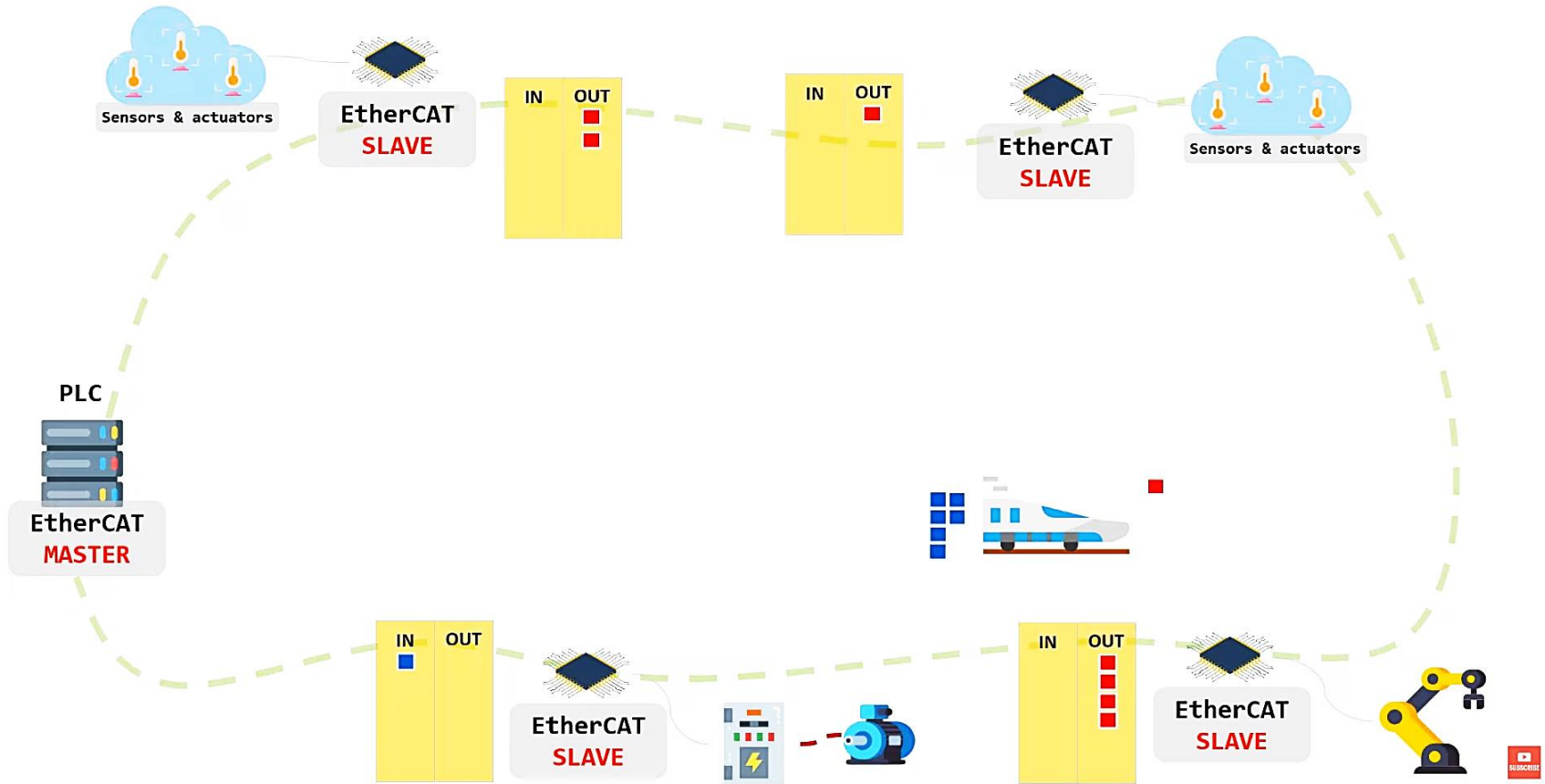
EtherCAT



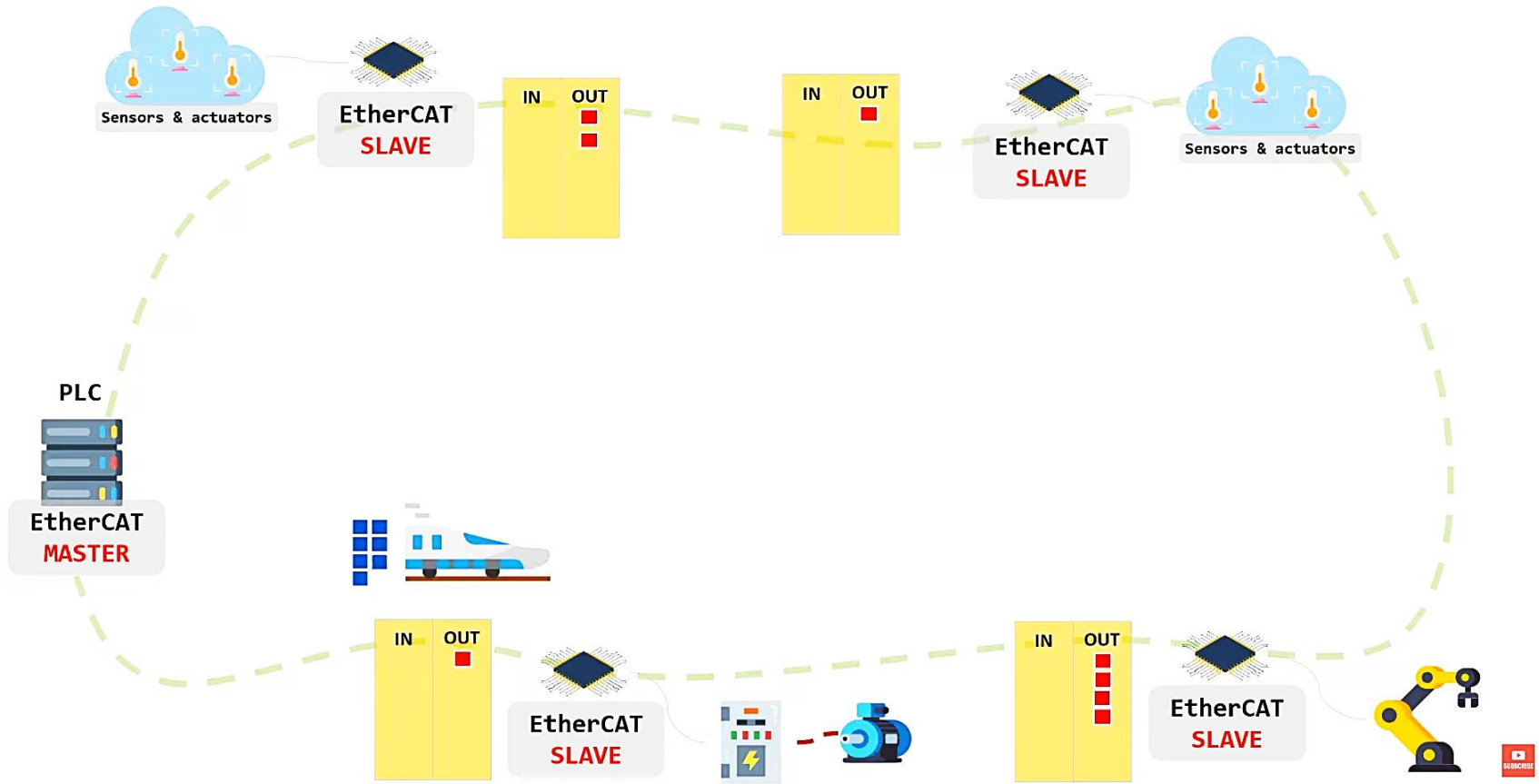
EtherCAT



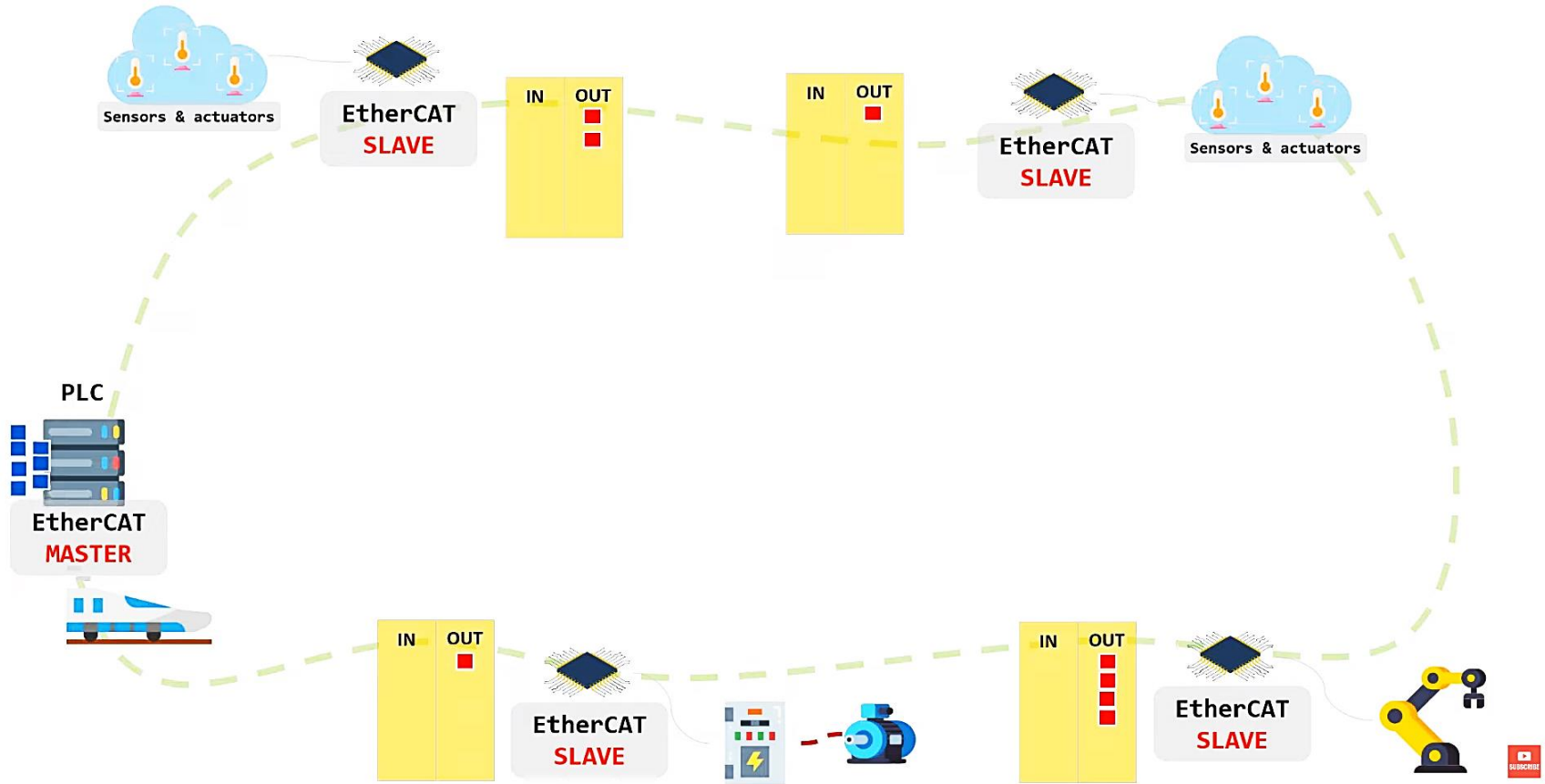
EtherCAT



EtherCAT

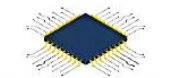
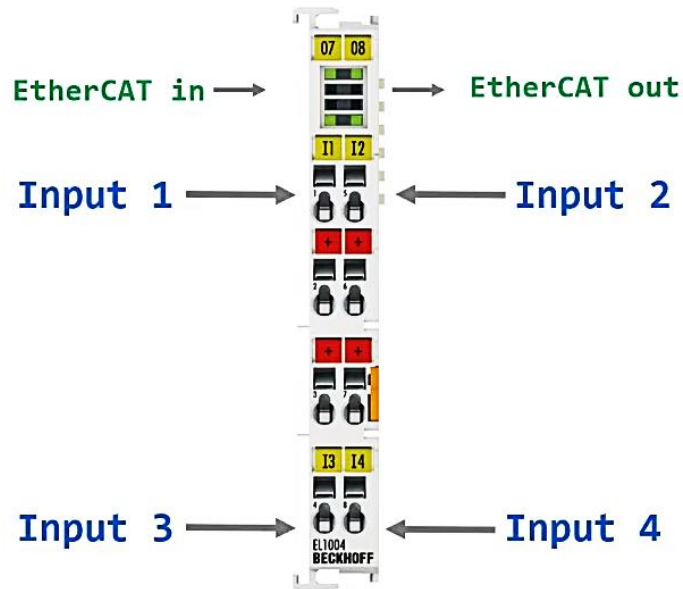


EtherCAT



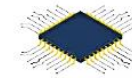
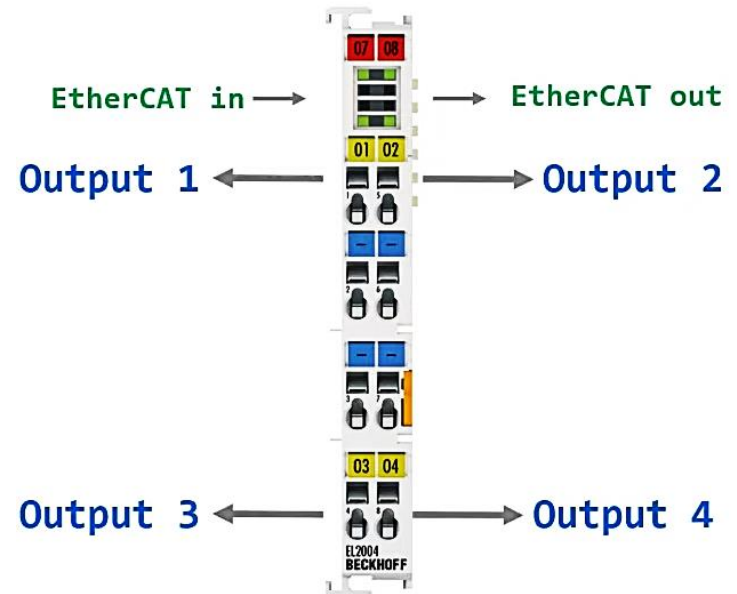
Dispositivos I/O

Beckhoff EL1004 terminal
4 digital **inputs**



EtherCAT
SLAVE

Beckhoff EL2004 terminal
4 digital **outputs**

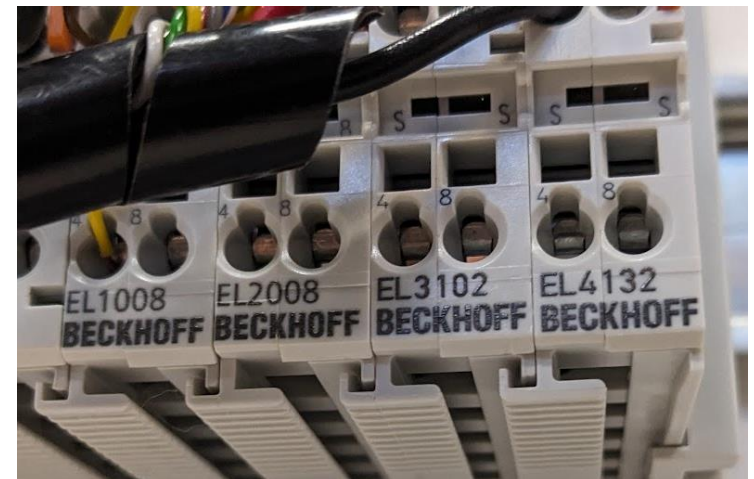


EtherCAT
SLAVE



Dispositivos I/O

- Ex10xx, entradas digitales
- Ex20xx, salidas digitales
- Ex30xx, entradas analógicas
- Ex400x, salidas analógicas

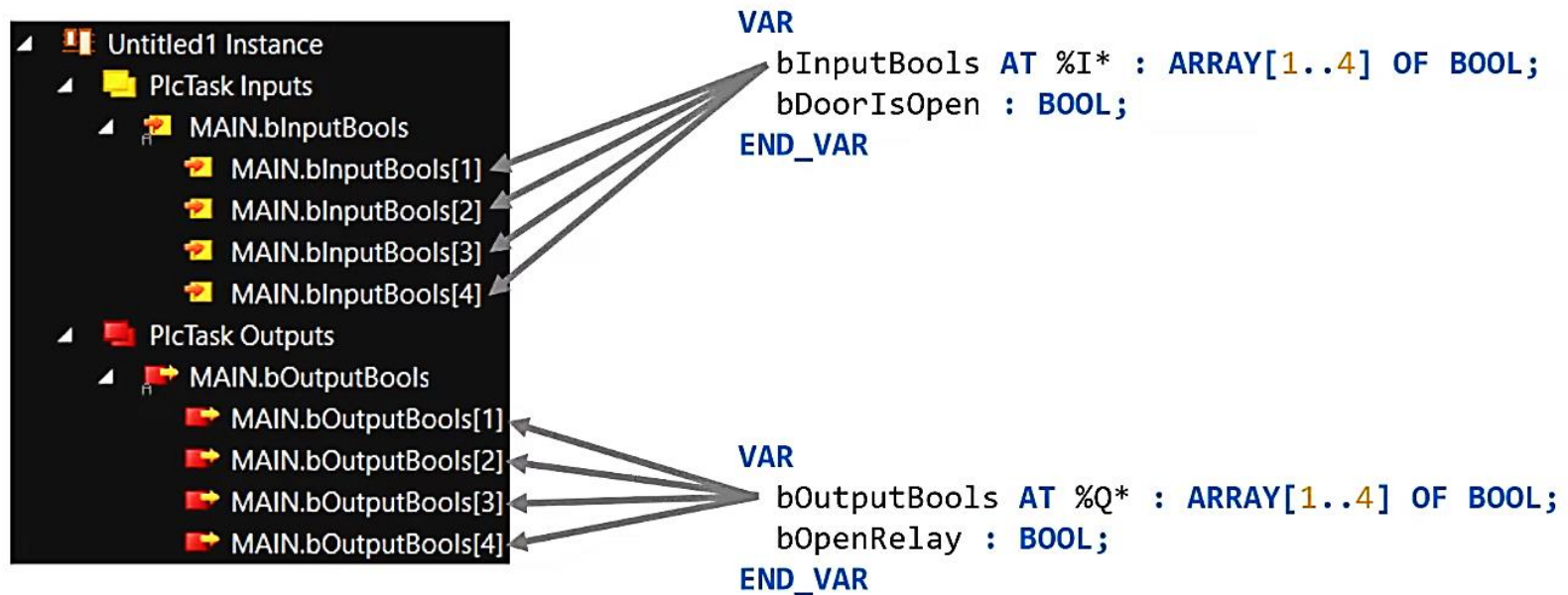


TwinCAT3 / Pasos

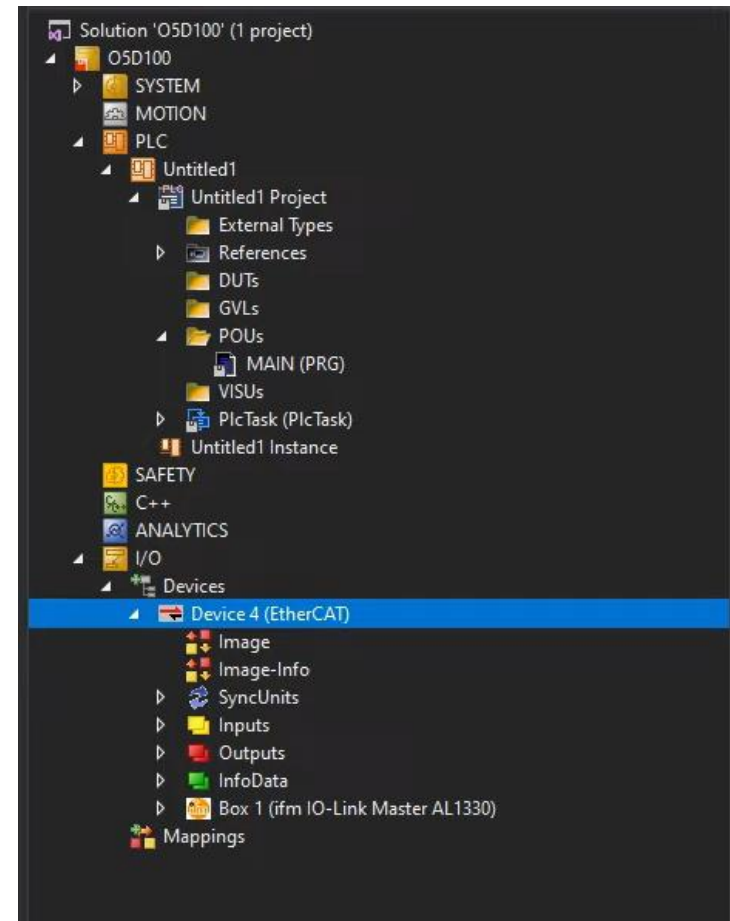
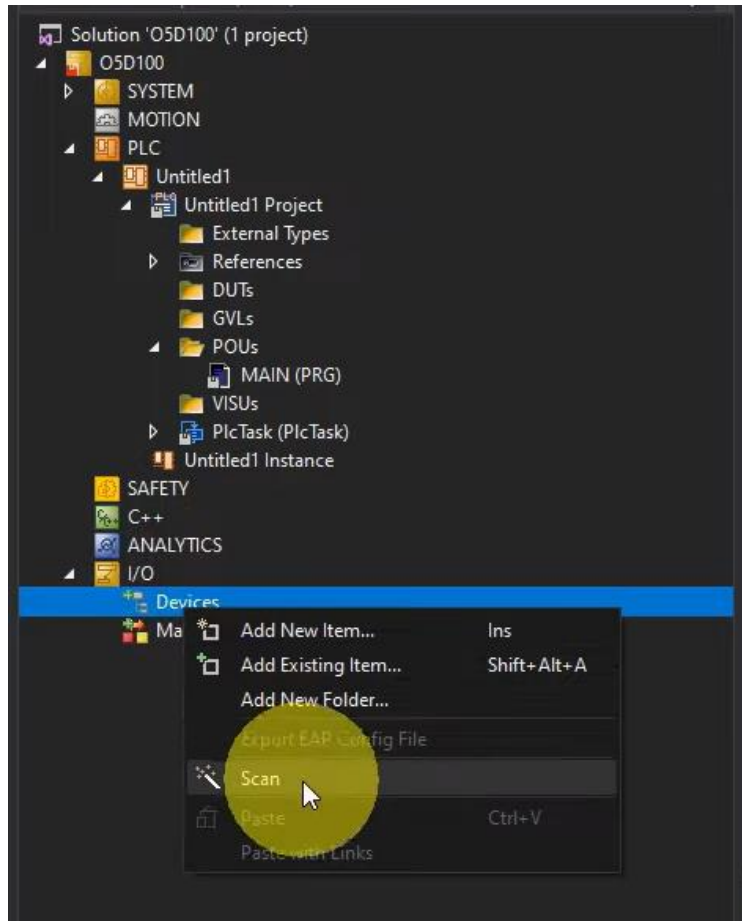
1. Declarar (variables)
2. Escanear (Config)
3. Compilar (hace visibles variables al IDE)
4. Enlazar (asignar variables a entradas/salidas)
5. Comprobar (modo RUN)

*modo CONFIG

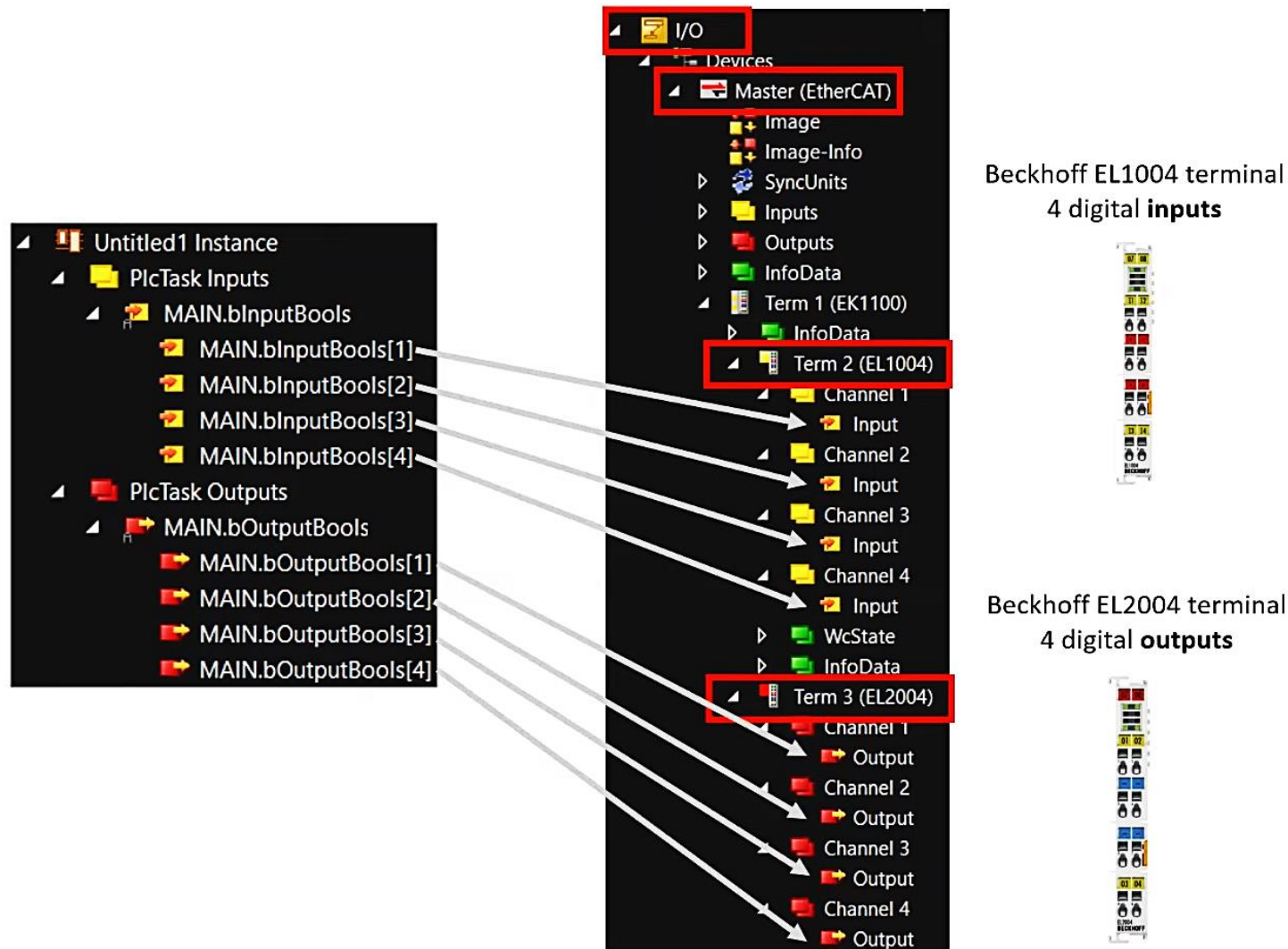
TwinCAT3 / Declarar



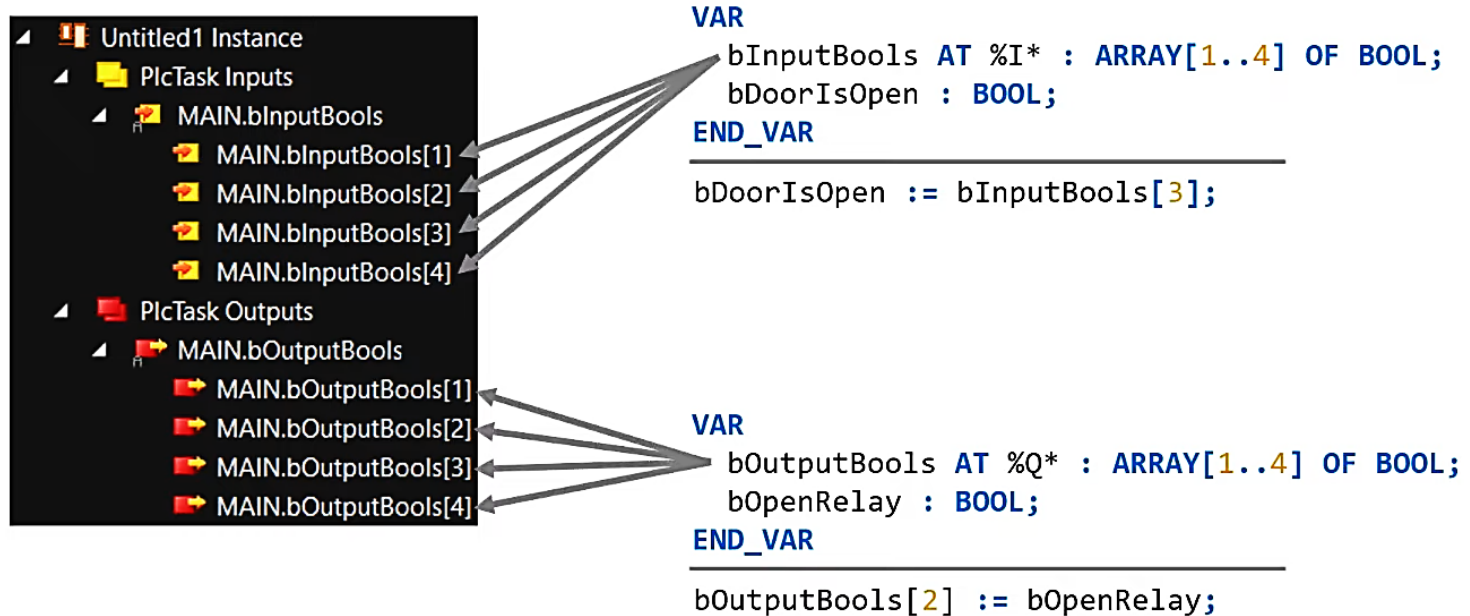
TwinCAT3 / Escanear



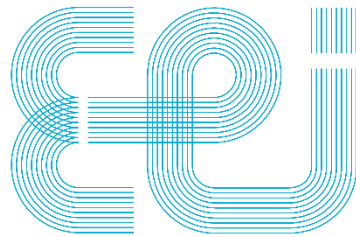
TwinCAT3 / Enlazar



TwinCAT3 / Uso



¿PREGUNTAS?
andsuarez@uvigo.gal



El contenido de esta presentación está basado en los materiales ofrecidos por
Jakob Satowski en <https://www.sagatowski.com/>